

UDCT v5.22

Audited Three-Dimensional Static-Field Note for Bullet-Cluster-Like Baryonic Offsets

Explicit QUMOND operator, Newtonian control, AQUAL cross-check, resolution audit, and
literature-corrected scope

Won Shik Paik
Independent Researcher, Auckland, New Zealand
22 September 2026

Status. This is a methods-correction and reproducibility note. It does not claim a Bullet Cluster fit, a new MOND mechanism, a relativistic lensing solution, or evidence that UDCT outperforms MOND or dark-matter models.

Version genealogy. v5.22 is the cluster-track successor to v5.21. It supplements, and does not replace, the locked v5.4 RAR, v5.5 BTFR, or v5.6 dwarf records.

Abstract

UDCT v5.21 replaced a local one-dimensional merger proxy with three-dimensional nonlinear field calculations and reported galaxy-dominant sampled contrasts in a favorable toy geometry. The present audit implements the required controls. The QUMOND interpolation and field equation are stated explicitly; Newtonian baryons are evaluated with the identical geometry, projection, sampling, and smoothing; the diagnostic is renamed from a peak ratio to a sampled galaxy-to-gas contrast; true global maxima are reported separately; grid spacing, box size, numerical floors, and affected-cell fractions are disclosed; and the result is placed against prior MOND Bullet Cluster calculations. At 100 kpc smoothing in the favorable toy geometry, the sampled contrasts are 1.328 for Newtonian baryons, 1.210 for QUMOND, and 1.160 for AQUAL. All three fall below unity by 150 kpc within the recomputed grids. Thus the toy galaxy-dominant contrast is not evidence of a uniquely nonlinear modified-gravity displacement: compactness, projection, and smoothing already produce it in the Newtonian baryonic control, while the modified operators reduce the contrast. QUMOND classifications remain stable over the tested 23.4-46.9 kpc grid spacings, but these grids do not establish continuum convergence. The strongest defensible conclusion is narrower: the old local proxy could not exclude a three-dimensional static route, yet the corrected toy experiment neither explains the observed Bullet Cluster lensing map nor supplies novelty beyond the recent literature.

Keywords: AQUAL; QUMOND; Bullet Cluster; Newtonian control; resolution audit; sampled contrast; reproducibility.

1. Frozen law and explicit field equations

The previously frozen acceleration scale is retained without cluster fitting:

$$g = g_N \sqrt{1 + \xi_0/g_N}, \quad \xi_0 = 1.082401 \times 10^{-10} \text{ m s}^{-2}.$$

The AQUAL interpolation function and equation are

$$\begin{aligned} \mu(y) &= 2y/[\sqrt{1+4y^2}+1], \quad y=|\text{grad } \Phi|/\xi_0, \\ \text{div}[\mu(|\text{grad } \Phi|/\xi_0) \text{grad } \Phi] &= 4\pi G\rho_b. \end{aligned}$$

The QUMOND comparison now has a fully specified interpolation and operator:

$$\text{laplacian } \Phi = \text{div}[\nu(|\text{grad } \Phi_N|/\xi_0) \text{grad } \Phi_N], \quad \nu(x)=\sqrt{1+1/x},$$

with laplacian $\Phi_N = 4\pi G\rho_b$. The high-acceleration limit of the frozen algebraic law is $g = g_N + \xi_0/2 + O(\xi_0^2/g_N)$, leaving a constant leading residual of $5.41 \times 10^{-11} \text{ m s}^{-2}$. Cluster cores can lie closer to this limit than outer galaxy regions.

2. Geometry and corrected diagnostics

The audit retains the two deliberately simple Plummer configurations. The favorable case was selected after an exploratory scan and is therefore an existence example, not a prediction.

Quantity	Baseline	Favorable
Total baryonic mass	$1.0 \times 10^{14} M_{\text{sun}}$	$3.0 \times 10^{14} M_{\text{sun}}$
Gas mass fraction / scale	0.85 / 300 kpc	0.85 / 500 kpc
Each galaxy fraction / scale	0.075 / 50 kpc	0.075 / 70 kpc
Galaxy coordinates	+/- 500 kpc	+/- 300 kpc
Status	Original baseline	Post-hoc favorable toy

Table 1. Toy inputs. These are not a reconstruction of the observed baryonic map.

The v5.21 quantity called a peak ratio was not the ratio of global maxima. It sampled the smoothed projected diagnostic density at three preselected coordinates:

$$C_{\text{samp}} = [\Sigma_{\text{off}}(-d,0) + \Sigma_{\text{off}}(+d,0)] / [2\Sigma_{\text{off}}(0,0)].$$

It is now called the sampled galaxy-to-gas contrast. The global argmax of the complete smoothed map is reported independently. A value above unity does not by itself reproduce an observed lensing peak, because the coordinates, source model, smoothing kernel, and lensing prescription are not observationally locked.

3. Newtonian control changes the interpretation

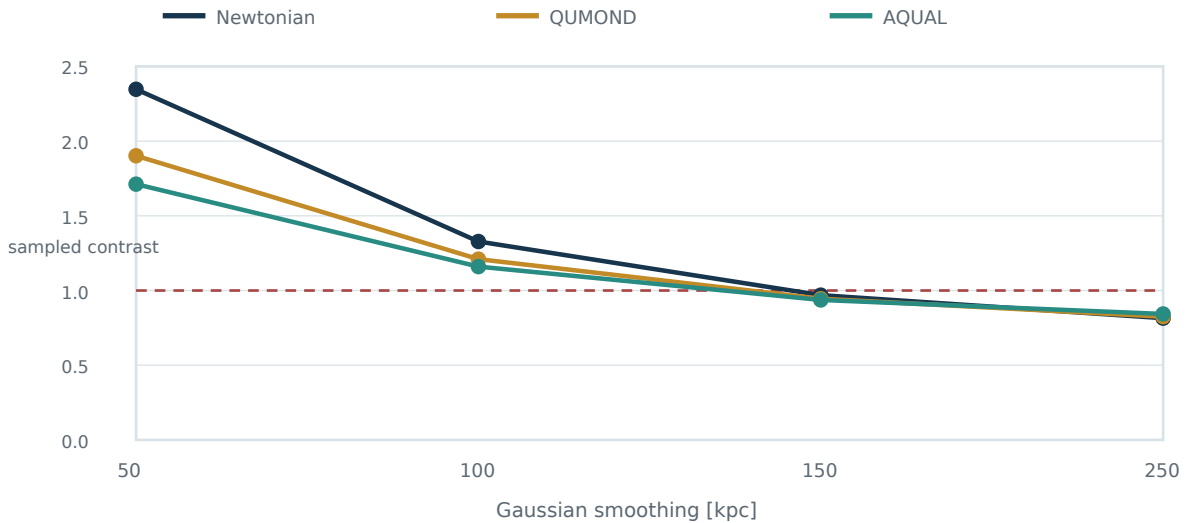


Figure 1. Recomputed favorable-geometry sampled contrasts. Newtonian and QUMOND use $N=128$ and a 3000 kpc box ($\Delta x=23.44$ kpc). AQUAL uses $N=96$ in the same box ($\Delta x=31.25$ kpc). The dashed line is $C_{\text{samp}}=1$.

Smoothing	Newtonian	QUMOND	AQUAL	Global argmax class
50 kpc	2.347	1.902	1.712	Near galaxy (+/-300 kpc)
100 kpc	1.328	1.210	1.160	Offset toward galaxy (argmax ~250 kpc)
150 kpc	0.969	0.945	0.937	Intermediate, nearer center
250 kpc	0.815	0.825	0.843	Gas center

Table 2. Favorable toy audit. The Newtonian control is larger than either modified-gravity result at 50-150 kpc.

Newtonian and QUMOND contrasts are from $N=128$, $\Delta x=23.44$ kpc; AQUAL is from $N=96$, $\Delta x=31.25$ kpc in the same 3000 kpc box. Classification can be compared; the 1.210 versus 1.160 difference is not a precision ranking of the two operators.

Table 4 is taken from the locked QUMOND CSV, not from a mixed earlier sweep.

4. What the control permits and forbids

The favorable toy geometry produces a galaxy-dominant sampled contrast even before a nonlinear modified-gravity operator is applied. The effect therefore cannot be attributed uniquely to AQUAL or QUMOND. Its primary ingredients are the compact galaxy components, diffuse gas, line-of-sight projection, and the chosen smoothing scale. In this setup the nonlinear operators redistribute the diagnostic density and reduce the galaxy-to-gas contrast relative to Newtonian baryons.

Withdrawn v5.21 wording: "genuine nonlinear spatial effect" and any implication that three-dimensional nonlinearity newly creates the galaxy-side dominance.

Retained statement: a local or one-dimensional proxy was not sufficient to decide the outcome of a multi-center field problem.

The local MOND boost is larger in the lower-acceleration diffuse gas than near the compact galaxy cores. The galaxy-side sampled dominance is therefore not evidence that the galaxy component is "more MONDian." It arises from compactness plus projection and smoothing, with nonlinear coupling superposed.

5. Numerical audit

Solver / case	N	Box [kpc]	Δx [kpc]	Floor	Affected cells
QUMOND, both	96	3000	31.25	$x=10^{-12}$	Same order as other QUMOND rows
QUMOND, both	128	3000	23.44	$x=10^{-12}$	3.81×10^{-6}
QUMOND, both	128	6000	46.88	$x=10^{-12}$	3.81×10^{-6}
QUMOND, both	160	6000	37.50	$x=10^{-12}$	1.95×10^{-6}
AQUAL, baseline	96	3000	31.25	$\mu: y=10^{-5}$	0
AQUAL, favorable	96	3000	31.25	$\mu: y=10^{-5}$	0

Table 3. Disclosed grid and regularization information. QUMOND floor fractions are dominated by exact or near-zero-field cells; AQUAL used no floor-affected cell in the final fields.

Geometry / QUMOND grid	50 kpc	100 kpc	150 kpc	250 kpc
Baseline, $96^3 / 3000$	0.860	0.543	0.485	0.541
Baseline, $128^3 / 3000$	0.857	0.545	0.491	0.550
Baseline, $128^3 / 6000$	0.867	0.576	0.526	0.590
Baseline, $160^3 / 6000$	0.871	0.585	0.545	0.618
Favorable, $96^3 / 3000$	1.871	1.195	0.933	0.816
Favorable, $128^3 / 3000$	1.902	1.210	0.945	0.825
Favorable, $128^3 / 6000$	1.838	1.223	0.982	0.875
Favorable, $160^3 / 6000$	1.846	1.204	0.962	0.857

Table 4. QUMOND sampled contrasts across the audit matrix. The success/failure classification is stable, but the ratios vary at the several-percent level.

6. Resolution and boundary limitations

The AQUAL solutions converged in 12 Picard iterations, with final relative updates 1.70×10^{-5} (baseline) and 1.23×10^{-5} (favorable); all conjugate-gradient solves returned normally. Their 100 kpc contrasts are 0.568 and 1.160. The true global argmax is at the gas center for every baseline smoothing. In the favorable AQUAL map it is at $x=281, 250, 156$, and 0 kpc for smoothing of 50, 100, 150, and 250 kpc, respectively.

These are solver-convergence statements, not continuum-convergence statements. At $\Delta x=23\text{--}47$ kpc, the 50–70 kpc galaxy Plummer radii span only about one to three cells. The $N=128$ and $N=160$ checks show that the threshold classification is stable; they do not demonstrate that C_{samp} or the argmax position has converged. A publication-grade morphology test requires adaptive or much finer grids, a documented isolated solver, and convergence of the projected map itself.

The QUMOND audit uses periodic FFT Poisson solves. The AQUAL calculation uses an isolated monopole boundary on a finite cube. Their agreement in broad classification is a useful cross-check, but their boundary conditions are not identical. Box-size dependence is visible and must not be described as negligible.

The favorable gas scale of 500 kpc is more extended than the Brownstein-Moffat King core radius $r_c=278$ kpc, while the toy galaxy components are much smoother than individual BCG light profiles. This pairing makes a galaxy-coordinate contrast easier to obtain. The fact that the baseline remains gas-dominant is therefore an important counterweight, not a minor detail.

7. Prior literature and novelty boundary

The qualitative mechanism is not new. Angus, Famaey, and Zhao (2006) constructed multi-center MOND/TeVSe models with Bullet-like lensing morphology. Angus, Shan, Zhao, and Famaey (2007) concluded that collisionless residual mass remained necessary even in MOND. Most importantly, Hernandez (2026) used a 219-member galaxy catalogue, an X-ray gas model, and a full 3D QUMOND calculation; the predicted galaxy-centered morphology was qualitatively similar to the lensing reconstruction, while a baryonic mass increase of about 1.6 was still required. Famaey (2026) independently emphasized the residual cluster missing-mass problem using the updated JWST lens model.

Novelty claim retained: within the UDCT record, v5.22 corrects an overstrong internal rejection, supplies an AQUAL-QUMOND-Newtonian matched comparison, and records the falsification boundary.

Novelty claim rejected: discovery that compact galaxies can dominate a MOND/QUMOND effective-density or lensing morphology in the Bullet Cluster.

8. Observational quantities must not be conflated

The recent JWST model finds the subcluster halo aligned with its BCG to 4^{+4}_{-2} kpc, while the main component is elongated and double-peaked. The frequently quoted ~ 47 arcsec separation corresponds to roughly 207 kpc at $z=0.296$ and must not be treated as the same measurement as the $\sim 150\text{--}160$ kpc gas-to-galaxy offsets.

No observed Bullet Cluster map is fitted here. A future test must use an independent WCS-calibrated gas surface-mass map and a stellar mass catalogue constructed without using the target lensing convergence map. Otherwise the target leaks into the predictor.

9. Lensing limitation

The projected quantity used here is Σ_{eff} from $\rho_{\text{eff}} = \text{laplacian } \Phi / (4\pi G)$. It is a Newtonian-equivalent source diagnostic of the solved potential, not a complete photon-deflection prediction. In a metric theory the lensing potential depends on both scalar metric potentials, $\Phi_{\text{lens}} = (\Phi + \Psi)/2$. AQUAL alone does not specify the gravitational slip or a covariant photon sector. Hence a successful argmax would still not constitute a Bullet Cluster lensing solution.

10. Prospective test protocol

Item	Required lock before target comparison
Gravity law	Frozen ξ_0 , explicit μ and v ; no cluster refit
Inputs	Independent gas and stellar mass maps with WCS
Geometry	Documented deprojection and uncertainty ensemble
Numerics	Adaptive/fine grid; map-level box and resolution convergence
Smoothing	Published reconstruction kernel, fixed in advance
Outputs	C_samp, full-map argmax, aperture mass, morphology residuals
Controls	Newtonian baryons, QUMOND, AQUAL, and literature baseline
Failure rule	No post-hoc geometry or smoothing chosen to cross C_samp=1

Table 5. Minimum preregistration needed for an observational claim.

11. Conclusions

#	Audited conclusion
1	The old local/1D proxy was insufficient to reject all 3D static multi-center calculations.
2	The favorable toy has galaxy-dominant sampled contrasts at 50-100 kpc in Newtonian, QUMOND, and AQUAL calculations.
3	Because the Newtonian control is stronger, this result is not evidence for a uniquely nonlinear modified-gravity displacement.
4	The classification is stable across the tested grids, but neither morphology nor amplitude is continuum-converged.
5	The result is anticipated by published MOND/QUMOND work and does not solve the residual cluster mass or relativistic lensing problems.
6	A defensible next step requires independent observed baryons, a locked smoothing kernel, finer numerics, and a covariant lensing prescription.

Final lock sentence. UDCT v5.22 preserves only the methodological correction: a local one-dimensional proxy cannot exclude a three-dimensional static field. The present toy calculations do not establish a novel nonlinear offset, a Bullet Cluster fit, or an advantage over MOND; their galaxy-side contrast is already produced more strongly by the matched Newtonian baryonic control.

Reproducibility record

Files used for this audit: v522_audit_recompute.py, aqual_3d_compare.py, v522_qumond_newton_audit.csv, v522_aqual_baseline_N96_B3000.json, and v522_aqual_favorable_N96_B3000.json. The QUMOND audit uses periodic FFT derivatives; the AQUAL solver uses Picard iteration, conjugate gradients, face-centered coefficients, under-relaxation, and an isolated monopole boundary. Companion code and locked outputs: https://github.com/wspaik5/UDCT_v5.22_3D_static_field_audit. The CSV/JSON are the source of truth for Table 4.

References

- [1] Bekenstein, J. & Milgrom, M. (1984). Does the missing mass problem signal the breakdown of Newtonian gravity? *ApJ* 286, 7-14.
- [2] Milgrom, M. (2010). Quasi-linear formulation of MOND. *MNRAS* 403, 886-895. doi:10.1111/j.1365-2966.2009.16184.x.
- [3] Angus, G. W., Famaey, B. & Zhao, H. S. (2006). Can MOND take a bullet? *MNRAS* 371, 138-146. arXiv:astro-ph/0606216.
- [4] Angus, G. W., Shan, H. Y., Zhao, H. S. & Famaey, B. (2007). On the Proof of Dark Matter, the Law of Gravity, and the Mass of Neutrinos. *ApJ* 654, L13-L16. arXiv:astro-ph/0609125.
- [5] Clowe, D. et al. (2006). A Direct Empirical Proof of the Existence of Dark Matter. *ApJ Letters* 648, L109-L113.
- [6] Brownstein, J. R. & Moffat, J. W. (2007). The Bullet Cluster 1E0657-558 evidence shows modified gravity in the absence of dark matter. *MNRAS* 382, 29-47.
- [7] Hernandez, X. (2026). A detailed MOND modelling of the Bullet Cluster. arXiv:2604.10811.
- [8] Famaey, B. (2026). On the residual missing mass of the Bullet Cluster. arXiv:2605.10022.
- [9] Rihtarsic, G. et al. (2026). Mapping dark matter in the Bullet Cluster using JWST imaging and spectroscopy. arXiv:2601.22245.